

Grade 10 Maths book: ingestion in numbers

How the whole of Siyavula *Everything Maths* Grade 10 became a curriculum graph and a question bank for the American-curriculum Grade 10 course (course:us-g10-math-en): the time, the AI agents, the cost, what came out, how it was checked, and what is still to do.

Dated 2 October 2026

App version v0.11.0 (latest release)

Report rev 1

Design authority: Data Spine thesis v1 (ADR-0001)

Not live yet: local pilot database only

The short answer

- Yes, the whole book is through. All 14 chapters went through the pipeline and passed the final gate (G5): 13 chapters with written holds, Chapter 11 clean. The drift guard (parity) is green.
- What came out: 211 learning objectives across 65 lessons, joined by 294 prerequisite links, and **5,326 live questions** (2,090 from the book, 3,137 generated variants, 99 interactive widget questions), plus 300 misconception explanations.
- What it took: about **\$943 of AI compute** (API list price), **4,394 AI agents** in 280 pipeline runs, about 1.2 billion tokens. Inside the ≈ \$0.85–1.1k approved for the book.
- How long: an 18½-hour pilot on Chapter 8 (26 Sep), then the other 13 chapters in **24 hours** of wall-clock time (1–2 Oct).

Read before sharing further. This content is **not in front of students**. It sits in a local pilot database; it is on a feature branch, not on (main) and nothing is deployed. Apart from Chapter 8, every quality gate was passed by the AI line, with each decision recorded. **No person has yet read the live questions:** human review in the admin console is the next job.

AT A GLANCE

Twelve numbers

14 / 14
chapters through all five gates

497
printed pages read

65
lessons

211
learning objectives

294
prerequisite links between objectives

5,326
live questions (book + generated + widgets)

300
misconception explanations

8,549
maths images turned into typeset maths

\$943
AI compute, metered (a floor; see notes)

4,394
AI agents in 280 pipeline runs

24 h
wall clock for 13 chapters in parallel

102
wrong printed answers found in the book

TIMELINE

From decision to full book in eight days

25 Sep 2026
Curriculum tracks decided
Students pick a curriculum at sign-up; the console gates content per curriculum and grade. First course: Grade 10 American maths, from the Siyavula book.

26 Sep · 04:38–23:04 UTC
Pilot: Chapter 8 (Analytical geometry) end to end
Every stage run once on one chapter to measure cost and quality before scaling. Samuel personally decided its objective gate (G1) and question gate (G2). Its measured cost re-based the estimate for the whole book.

27–30 Sep
Pipeline hardened from the pilot's findings
What the pilot exposed was fixed, and the rules it raised were written down as decisions from Samuel's answers.

1 Oct · 07:56 UTC
Go for the full book; fan-out starts on chapters 1–7 and 9–14 in parallel
Samuel's answer 37: run the whole book, let the gates auto-pass, and put every decision in the console for review (budget ≈ \$0.85–1.1k). The maths reading for the whole book, then lessons, questions, checks, misconceptions, generated questions and widgets, chapter by chapter.

1 Oct · evening
A defect found mid-run and repaired
Worked-example step titles had lost their maths ("Divide both sides by ...") with the expression missing: 138 of 571 titles. Fixed at the source and repaired in the saved runs and the database without new model calls.

2 Oct · 07:55 UTC
Last chapter through G5; drift guard green
All 14 chapters have a G5 record. The status record and the errata list were written up the same morning.

Next
Human review, merge, release
See "What is not done yet" below.

TIME

Hours spent

18.4 h
pilot, Chapter 8, wall clock (26 Sep)

24.0 h
full-book fan-out, wall clock (1–2 Oct)

26.1 h
summed run time of the fan-out's workflows (they overlap)

≈ 21 h
active orchestration time on G10 (the coordinating Claude session)

Human time is not metered. Samuel's part was decisions, not labour: **45 written answers** that steer the pipeline (recorded in (samuel-answers.md) and the spec's decisions table), and his own G1 and G2 on the Chapter 8 pilot. The coordinating session ran about 55 active hours since 20 Sep in all; the rest went to other work (console, releases).

AI EFFORT AND COST

Where the \$943 went

Every pipeline run is metered to a ledger ((runs/g10-math/cost.json)) at API list prices as of 1 Oct 2026. These are API-equivalent figures, not an invoice.

By stage

Maths reading
50B: equation images → typeset maths
Step-by-step working check
SW: every solution checked twice
Generated question families
56: author, then grade
Misconceptions
55: draft, then final
Lessons and book questions
52–54
Learning objectives
51: finders + reconciler
Interactive widgets
57: author, then blind verify
Question-gate recommendations
G2B: doubtful items re-judged
Calibration
50B sample run

\$285.94 · 322 ag.

\$158.71 · 1,301 ag.

\$134.65 · 853 ag.

\$122.79 · 551 ag.

\$109.70 · 722 ag.

\$73.51 · 378 ag.

\$38.72 · 146 ag.

\$14.77 · 117 ag.

\$4.53 · 4 ag.

Total: \$943.32 · 4,394 agents · 280 runs · 1,197.5 million tokens (982M re-read from cache, 185M written to cache, 30M generated).

By model

Sonnet 5.5
Sonnet 5.5 \$726
Sonnet 5 \$200
Haiku 4.5 \$17

Every pipeline agent ran on Sonnet or Haiku. Opus was kept for the coordinator. The run moved to Sonnet 5.5 on 1 Oct (Samuel's answer 41).

Unit economics

per live question
per printed page
per learning objective
per lesson
per chapter, excluding the one-off maths reading

\$9.18
\$1.99
\$4.47
\$14.51
\$46.57

One-off cost for a book. Serving costs (tutoring turns) are separate and metered per student.

By chapter

The book-wide maths reading (\$290.47, 326 agents) is not split by chapter. Chapter 8 includes the pilot's learning runs.

1: Algebraic expressions
2: Exponents
3: Number patterns
4: Equations, inequalities
5: Trigonometry
6: Functions
7: Euclidean geometry
8: Analytical geometry (pilot)
9: Finance and growth
10: Statistics
11: Trigonometry
12: Euclidean geometry
13: Measurements
14: Probability

\$80.08 · 592 ag.
\$24.46 · 186 ag.
\$16.55 · 109 ag.
\$47.58 · 341 ag.
\$48.45 · 291 ag.
\$109.46 · 647 ag.
\$46.32 · 246 ag.
\$114.59 · 583 ag.
\$31.03 · 225 ag.
\$34.98 · 219 ag.
\$13.50 · 66 ag.
\$8.06 · 48 ag.
\$34.34 · 225 ag.
\$42.52 · 286 ag.

Chapters sum to \$651.93 and 4,064 agents; with the maths reading (\$290.47, 326) and one book-wide run (\$0.92, 4) the total is \$943.32 and 4,394.

How exact is \$943?

It is a floor, within about 1%. (1) The ledger file has 284 rows totalling \$952.70; four of them repeat a run already counted, so 280 unique runs and \$943.32. (2) Five runs are marked incomplete, and the maths-reading and working-check stages include some tokens the meter could not price. (3) **Not included:** the coordinating Claude sessions themselves. Estimated at list price, the main session (≈ \$1.1k) and its 169 helper agents (≈ \$3.35k) cover *all work* since 20 Sep, not only this book, so they are not a G10 cost and are not split here. Actual billing depends on the plan in use.

INPUT

What went in: the book

497
printed pages, 14 chapters

4,599
content blocks read from the e-book

8,549
maths images transcribed and accepted

964
figure images extracted (462 served as pictures)

The blocks, by kind

Exercise items	Paragraphs	Summary items	Headings	Worked examples	Boxes	Figures	Definitions
2,531	696	219	197	174	136	75	47

The other 524 blocks are of smaller kinds. The book prints maths as images; each one was read by a model into typeset maths and accepted only after a check. Until native figures are drawn, the book's own pictures stand in for figures (Samuel's answer 37d).

OUTPUT 1

What came out: the curriculum graph

The graph is the product's spine: objectives a student can master, the lessons that teach them, and which objective must come before which.

14
modules (one per chapter)

65
lessons

211
learning objectives

294
prerequisite links

1,964
teaching claims, each cited to the book

358
worked-example entries

616
visuals attached

300
misconception explanations

For scale: the Prep-3 maths book on the site today has 10 modules, 90 objectives, 112 prerequisite links and 450 book questions. Grade 10 has **2.3×** the objectives, **2.6×** the links and **4.8×** the book questions (2,154), before counting any generated variants.

OUTPUT 2

What came out: the questions

5,326 live questions

Book · 2,090
Generated variants · 3,137
Interactive widgets: 99

"Live" means ready to serve once the course is switched on. A further 6A book questions are on automatic hold (the sources disagree, or the answer cannot be marked safely), 41 were rejected (book error or damaged text) and 4 widget questions were retired. Generated questions are new numbers on a book question's pattern, never new maths: 336 families were authored, 314 passed grading.

2,700 book items parsed, by answer type

Expression
Number
Not markable
Choice

1,284
868
339
289

2,526 exercises and 174 worked examples. "Not markable" means a proof, sketch or explanation that cannot be marked automatically.

2,154 bank questions, by difficulty

Basic
Standard
Advanced

704
1,231
219

Every one carries the book's own worked solution (2,047 from the e-book's worked answers, 107 from its worked examples). The tutor explains from these and never solves from scratch: grounded teaching.

QUALITY

How it was checked

Five gates stand between the book and a student. Each records its decisions and reasons so a person can audit them in the console. On this run the AI line passed them (decisions 58c and 60); Samuel passed G1 and G2 himself on the pilot.

G1 Objectives. Is every objective really taught in the book, with evidence? Approve, edit or drop. 508 decisions

G2 Book questions. Do the printed answer, the e-book solution and a blind re-solve agree? Accept, reject or exclude. 700 decisions

G3 Generated questions and widgets. A sample of every family is read; a rejected family is retired. 600 decisions

G4 Misconceptions. Are dropped explanations really wrong, and are kept ones sound? 376 decisions

G5 Go / no-go. Coverage, drift guard, cost ledger, the change about to load. Result: 13 chapters pass with holds. Chapter 11 pass. 80 decisions

2,264 gate decisions in all, each written to a gate record.

Doubtful book questions re-judged

383 items where the sources disagreed went to a recommendation run; 206 recommendations were confirmed by an independent verifier.

accept 128
fix 94
hold 39
exclude 122

The book's own errors

102 printed answers look wrong, and 20 questions were damaged in extraction (chapters 1–7, 9–14).

A question with a wrong printed answer is left out, never silently corrected. The list (errata-g10.md) waits for a reader.

Step-by-step working check

Each of the 2,556 worked solutions was read twice, line by line, by a checker calibrated on the pilot (16 of 16 real errors found, no false alarms, about 5 cents a solution). It raised 300 flags on 270 solutions. A flag goes to the console's review list; it never changes the content.

Arithmetic slip
Wrong label or name
Other
Wrong value carried
Sign error
Copied wrongly
Final answer

65
65
53
42
34
28
13

Coverage: safe everywhere, not yet complete everywhere

Every safety check passes in every chapter (among them: the maths renders, notation is consistent, every question has a sourced solution, teacher-only text stays hidden). The completeness checks are what keep 13 chapters on "pass with holds":

- 178 of 211 objectives (84%) have full evidence; 33 lack a book question or a cited claim.
- 554 of 633 objective-difficulty slots (88%) have a live question; 79 are empty, mostly in chapters 5, 6 and 10.

Each gap is either signed off by a person as an exception or filled with new generated questions or widgets: an open decision for Samuel.

Defects the line caught in itself, and fixed

- Step titles lost their maths (138 of 571 worked-example titles): fixed at the source; runs, bundles and the database repaired with no new model calls.
- Coordinate notation: the book writes (x; y); students now see (x, y) everywhere they read text.
- A wrong sample space ((1,1),(1,2)... read as decimals) in one probability question, invisible to the notation check: found and fixed.
- Maths that would not render in Chapter 13 (47 strings): repaired at assembly.
- Drift guard now counts only servable questions (Samuel's answer 45); a one-objective lesson is allowed once with a "check later" note (answer 44).

DETAIL

Chapter by chapter

Read from the pilot database on 2 Oct 2026. Book questions: live / held / rejected. Generated: live variants (families passed of authored). Flags: step-check flags (solutions flagged).

Ch	Title	Lessons	Objectives	Links	Book questions	Generated	Widgets	Misc.	Flags	G5
1	Algebraic expressions	8	29	24	517 / 6 / 27	428 (43/45)	7	30	55 (53)	holds
2	Exponents	3	9	21	152 / 0 / 3	139 (14/15)	0	8	8 (8)	holds
3	Number patterns	1	5	6	75 / 1 / 2	140 (14/14)	0	4	8 (8)	holds
4	Equations and inequalities	6	19	30	259 / 6 / 9	250 (25/26)	20	18	33 (31)	holds
5	Trigonometry	5	18	30	210 / 8 / 0	150 (15/16)	6	23	31 (25)	holds
6	Functions	9	37	64	217 / 22 / 0	690 (69/78)	24	82	38 (38)	holds
7	Euclidean geometry	4	13	19	103 / 8 / 0	250 (25/27)	3	19	23 (20)	holds
8	Analytical geometry (pilot)	5	13	1	149 / 9 / 0	110 (11/11)	21	29	27 (25)	holds
9	Finance and growth	4	15	19	133 / 0 / 0	200 (20/22)	1	17	14 (13)	holds
10	Statistics	5	16	18	80 / 0 / 0	180 (18/20)	7	24	13 (11)	holds
11	Trigonometry	1	3	10	24 / 0 / 0	70 (7/7)	1	4	0 (0)	pass
12	Euclidean geometry	1	2	0	6 / 0 / 0	20 (2/2)	0	0	6 (6)	holds
13	Measurements	6	14	26	74 / 1 / 0	210 (21/21)	1	18	29 (18)	holds
14	Probability	7	18	26	91 / 3 / 0	300 (30/32)	8	24	15 (14)	holds
	Book	65	211	294	2,090 / 64 / 43	3,137 (314/336)	99	300	300 (278)	

Chapter 12 is a proof chapter: no misconceptions were produced and no prerequisite link survived the checker. Chapter 8 has one link and none to other chapters; both are open decisions. Four more Chapter 8 widget questions are retired.

ACCOUNTABILITY

Who decided what

Samuel (a person)

- Chapter 8 pilot: the objective gate and the question gate, 26 Sep.
- 45 written answers that set the rules: go for the full book with auto-passing gates, book pictures as stand-in figures, quality over cost, which model, how multi-part questions read.
- Two rule changes on 2 Oct: the drift guard counts servable questions only; one lesson may keep a single objective, marked "check later".

The AI line (agents)

- Every gate on chapters 1–7 and 9–14, under Samuel's auto-pass rule, each decision recorded with its reason.
- Six objectives dropped and their 78 questions moved to a neighbour, by an AI ruling that needs a standing policy.
- Fixes to its own defects, each tested and marked in the files; none yet reviewed by a person.

OPEN

What is not done yet

- Human review in the console. The live questions, generated variants, widgets, misconceptions, the 102 errata and the 300 working-check flags have not been read by a person. This is the next job.
- Book-level close-out. Chapter 8's assembled content is not yet on the branch; the whole-book export, the book's status and the book-wide coverage report still have to run before the course can be loaded anywhere else.
- Merge, release, deploy. The work sits on a branch forked at v0.9.3; (main) has since released v0.10.0 and v0.11.0. A trial merge shows 15 conflicts to resolve, then a pull request, green CI, a release and a deploy on Samuel's go.
- Refresh stale records. The G5 records of chapters 1–4, 8 and 14 predate later fixes; the misconception drafts of chapters 1–5 were built before the question-gate recommendations. Re-running them is cheap (≈ \$1.3 an objective for misconceptions) and is Samuel's call.
- One objective with nothing live (Chapter 8.1, "drawing figures from coordinates"): three question families are written and graded but not loaded.
- Paperwork. Requirement and traceability entries are owed for several rules the run introduced; three decisions (62–64) are missing from the table.

14
open decisions that need Samuel

7
content items waiting for a reader

8
engineering and records items

All 29 are listed with where to look in (docs/WIP-g10-pilot/README.md), section "Open decisions after the full-book run".

DESIGN AUTHORITY

How the pipeline was built: the Data Spine thesis

The ingestion line is not ad hoc. ADR-0001 makes The Agent-Native Data Spine — A Thesis (founder's thesis v1, June 2026; 20 chapters, about 75,000 words, 28 sources) the design authority for the solution architecture, and ADR-0005 builds the extraction pipeline on it. How its ideas show up in this run:

Pillar 1: schema first, not text first	Every stage writes typed records (objectives, claims, questions with an answer spec) that are validated before they load, rather than free text.
Ch. 2–3: document understanding and deterministic extraction	The e-book is read by structure (4,599 blocks), maths images become typeset maths with an acceptance check, and assembly is plain code with no model calls; this is why the step-title defect could be repaired by a script rather than by re-running the models.
Ch. 15: curriculum graphs and educational agents	Objectives, lessons and modules with a prerequisite graph (15.1 schema); every question and claim cites the objective and the book passage it came from (15.5).
Ch. 15.7: build a spine, sell verticals	A second curriculum's book arrives as a new course beside Prep-3 on the same graph and the same gates, switched on per curriculum and grade.
Ch. 16: bitemporal knowledge	Syllabus versioning on graph edges is the design (ADR-0001); this run loads one version and does not yet exercise it.
Ch. 19.6: MVP-cut discipline	One chapter piloted end to end, measured, then the rest fanned out; nothing is built that the pilot did not need.

VERSION AND SOURCES

Version

Report	rev 1 · 2 October 2026
App version	v0.11.0, the latest release on (main) (2 Oct 2026)

Where this work is branch (feat/003-curriculum-tracks-g10-american-math) at 961d691, forked from v0.9.3; not merged, not released, not deployed

Data as of pilot database and cost ledger, both read 2 Oct 2026

Prepared by Claude (the coordinating agent) for Samuel Toma, CTO. Figures come from records, not from a person's audit.

References

- The Agent-Native Data Spine — A Thesis. Founder's thesis v1, June 2026. (agentic-data-thesis.html)
- ADR-0001 Architecture follows the data-spine thesis. (docs/decisions/0001-architecture-follows-data-spine-thesis.md)
- ADR-0005 The extraction pipeline. (docs/decisions/0005-extraction-pipeline.md; spec (docs/specs/extraction-pipeline.md)
- Spine-derived architecture. (docs/architecture/spine-derived-architecture.md)
- Feature spec 003, curriculum tracks: spec, plan, decisions. (specs/003-curriculum-tracks/)
- Full-book status record, errata and Samuel's answers. (docs/WIP-g10-pilot/)
- Cost ledger and gate records. (services/extraction/zuns/g10-math/)
- Source book: Siyavula, *Everything Maths* Grade 10.